

ANIRUDH NEGI

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SUMMARY

Backend engineer with 8+ years building distributed systems and data platforms at scale: event-driven integration, graph and relational data modeling behind recommendation and search, and large-scale ETL across GoFundMe, Genentech, and SymphonyAI. I own architecture end-to-end, including schema and API contracts, failure semantics, migration strategy, and the cross-team alignment that makes them stick. Outside of work I architect and operate two live multi-service products built almost entirely through agentic AI workflows with Claude Code. I own the design, the reviews, and the reliability; the agents write most of the code.

CORE COMPETENCIES

Distributed Systems & Architecture: Event-driven architecture, service decomposition, API contract design, idempotency & exactly-once semantics, fault tolerance, backpressure & retry/DLQ strategy

Data Modeling & Retrieval: Relational and graph data modeling (Neo4j property graphs for recommendation), large-scale ETL, CDC & incremental load, search index design (Algolia, Elasticsearch/OpenSearch)

Production Migrations: Zero-downtime schema and provider migrations, dual-write/backfill/reconcile, idempotent + resumable batch, rollback design

AI-Native Engineering: Claude Code / agentic development workflows, LLM provider abstraction & failover, versioned prompt registries, MCP servers, structured-output validation, cost & latency telemetry

Cloud, Data & Languages: AWS (ECS, Lambda, RDS, S3), Kubernetes, Helm, Terraform, Docker, CI/CD · PostgreSQL, Neo4j, Redis, Greenplum · Kotlin, Python, TypeScript, Java, SQL · FastAPI, Spring Boot, React

PROFESSIONAL EXPERIENCE

GoFundMe · Software Engineer II, Backend & Platform

Mar 2023 to Present · Kotlin, TypeScript, AWS, Terraform, Kubernetes

- **Owned the architecture of the event-driven integration platform** connecting GoFundMe's distributed fundraising services to Salesforce Nonprofit Cloud, defining the event schema, delivery guarantees, retry/backoff policy, and dead-letter strategy that hold data consistent under partial failure. Replaced a tightly coupled sync path that generated recurring manual reconciliation work.
- Served as **primary technical point of contact across product, data engineering, and platform teams** for that integration, driving contract alignment and event semantics through design docs and review.
- **Modeled the platform's charity/fundraiser/donation domain as a Neo4j property graph and built the sync pipeline that keeps it current.** Designed the node and relationship schema powering a recommendation product, choosing a graph store over relational joins because the traversals that drive recommendations are the ones relational modeling makes most expensive.
- **Designed and led ETL pipelines spanning 1.8M+ charities and 6M+ fundraisers.** Built resilience in as a property of the design (idempotency keys, checkpointing, graceful degradation) so runs resume mid-window without loss or duplication; defined freshness SLAs with downstream consumers and instrumented lag, error rate, and throughput so degradation surfaced before it reached them.
- **Re-architected the Algolia search-index sync, cutting a 10+ hour run to 4 hours across millions of records** by restructuring the batch strategy and eliminating redundant work, so the index stays current within its operational window instead of overrunning it.
- **Shipped LLM-backed content generation into production fundraising workflows,** cutting manual content creation overhead across the platform; prototyped multi-step agentic orchestration for internal tooling and made the build-vs-buy call on frameworks.
- **Built prototypes and demos that turned proposed capabilities into decisions,** standing up working implementations so teams could evaluate ideas against real behavior rather than a design doc. Several were taken forward into the product.
- Built an asynchronous, non-blocking KYC/AML identity verification integration (Alloy) enforcing compliance gates without adding onboarding latency; authored reusable Terraform and Helm modules standardizing Kubernetes deployment. **SPOT Award** for cross-functional technical leadership.

Genentech (Roche) · Senior Java Developer

May 2022 to Mar 2023 · Java, Spring Batch, AWS, Terraform

- **Led a large-scale Salesforce-to-enterprise-database migration in a regulated environment,** designing the ETL, integrity validation, and reconciliation logic with idempotent execution and rollback so failed runs could be safely re-run without duplication or corruption. Replaced ad hoc scripts with a repeatable framework non-engineers could monitor via dashboards.
- **Cut end-to-end Spring Batch runtime by 60%** by profiling hot paths and restructuring chunk sizing, step parallelism, and reader-processor-writer I/O against memory and connection-pool limits, then documented the approach into the engineering

knowledge base for other teams to reuse. Provisioned the supporting AWS environments (EC2, RDS, S3, IAM) in Terraform, with plan review enforced before apply.

SymphonyAI · Software Developer

Feb 2019 to Jul 2021 · Java, Python, Elasticsearch

- **Decomposed monolithic sequential ETL jobs into concurrent, independently executable stages, taking end-to-end processing from hours to minutes**, partitioning load across threads and tuning partition size against hardware without overwhelming downstream systems.
- **Architected Elasticsearch indexing strategy, custom analyzers, tokenizers, and filter chains** for domain-specific and multilingual enterprise search, materially improving recall, precision, and ranking where standard analyzers failed. Ran cluster sizing and shard optimization that raised query throughput and reduced p95 latency under high concurrency.
- **Mentored junior and mid-level engineers** on distributed search and indexing design through walkthroughs and pair programming.

Mediaocean · Software Engineer

Feb 2016 to Jan 2019 · Java, SQL, Greenplum, Docker

- Built ETL pipelines consolidating enterprise master data from fragmented source systems into a Greenplum warehouse, giving analytics teams a single consistent source of truth. Designed incremental-load and CDC logic that shortened the nightly batch window, plus in-pipeline data quality checks that surfaced source anomalies before they silently corrupted output.
- Optimized the SQL behind live analytics dashboards via query-plan analysis, rewriting joins, eliminating full-table scans, and adding targeted indexes under high-concurrency reporting load. Containerized ETL workflows with Docker. **Rising Star** and **Brand Value** awards.

INDEPENDENT PRODUCTS · ARCHITECTED SOLO, BUILT WITH AGENTIC AI WORKFLOWS

Both deployed and solo-architected. I define the architecture, data model, and service boundaries and own reliability; Claude Code agents produce the bulk of the implementation under that spec, running as versioned scheduled skills that extend both products daily without supervision. Docs and packages: github.com/utilix-tech · npmjs.com/org/utilix-tech

Karyfy · AI job-search platform · karyfy.com (live, in beta)

FastAPI · React/TS · PostgreSQL · Redis · OpenSearch · Kubernetes/Helm

- **Architected a hub-and-spoke multi-service system**: a FastAPI hub exposing 250+ REST endpoints and owning identity, entitlement, and orchestration, fronting internal services for AI routing, resume curation, job extraction, coaching, career planning, and a shared identity schema. Chosen deliberately over separate public apps with independent SSO, so service boundaries stay internal and reversible.
- **Designed and built a dedicated AI Gateway** so no product service calls a model provider directly: provider abstraction over OpenAI and Anthropic with automatic failover, a **versioned prompt registry with a draft, canary, active, deprecated lifecycle**, tier-aware model routing under timeout budgets, structured-output schema validation with deterministic error codes, and per-request tracing and cost estimation. Product services keep a direct-LLM fallback path.
- **Designed the safety boundary that lets agents ship to main unattended**. Work is partitioned by label: a scheduled agent implements and merges auto-safe issues across every repo behind a green-test gate, while anything touching auth, payments, secrets, DB migrations, or deploys is fenced off as needs-human. A test-integrity rule forbids weakening a test to force green; the agent reverts and reports blocked instead. Content pipelines run on the same footing, authoring courses and posts into production on a 4-hour and weekly cadence.
- **Ran a full architecture audit and deliberately cut scope**: demoted an over-engineered Kubernetes/Helm setup to Compose for MVP, consolidated seven sibling products into modules, archived an unbuilt service.

Utilix · developer tools platform · utilix.tech

TypeScript monorepo (pnpm) · Next.js · Hono · Drizzle/Postgres

- **Designed a single shared tool registry that serves six delivery surfaces from one implementation**: a web app, a key-authenticated REST API, a Node SDK, a Python SDK, an embeddable widget, and an MCP server, so a tool is specified once and its contract holds identically everywhere instead of drifting across per-surface reimplementations. **180+ tools** and growing daily.
- **Ship as versioned public packages on a continuous release cadence**: `@utilix-tech/sdk` and `@utilix-tech/mcp` on npm, `utilix-sdk` on PyPI. **8K+ combined npm downloads/month**.
- **Built an MCP server exposing 130+ tools to AI agents** over stdio with Zod-validated schemas, giving Claude Code, Cursor, and any MCP client typed access to the full toolkit.
- **Automated the release pipeline end-to-end as a scheduled agent**: a daily routine researches, builds, tests, documents, and ships 1 to 2 new tools across every surface, then publishes to npm and PyPI with no human in the loop. The single-source-of-truth registry is what makes that safe, since one specification propagates to all six surfaces rather than six hand-edits drifting apart.

EDUCATION

Rutgers University · MS, Computer Science (Jan 2021 to Dec 2022) · Academic Excellence Award · Distributed systems, algorithms, machine learning, cloud computing.